Technical Appendix: An Empirical Analysis of Transfer for Dynamic Path Planning

Anonymous submission

A. Protocols by Stage

All continuous stages share the substrate: a 2-D point robot, maps with procedurally generated obstacles, a probabilistic roadmap (PRM) of exactly 192 nodes with k=7 connectivity, and A* evaluated at a per-suite *binding budget* of node expansions calibrated (on the anchor baseline only) so that baseline success falls in the pre-specified discriminative band. Unless noted, training labels are exact graph-Dijkstra costs on the PRM (static) or exact backward space–time Dijkstra values over (node, t) (dynamic). Headline continuous runs are local single-GPU (RTX 5090) validations with one primary training seed unless stated; the discrete program ran on a remote fleet.

A2. Exact Environment, Model, and Data Specifications

World generation. Continuous worlds are unit squares (side length 1.0; the large-room family uses 2.0); units are abstract. Start and goal are sampled with minimum separation 0.45 of the side length and inserted as roadmap nodes 0 and 1. Roadmaps contain exactly 192 sampled nodes with k=7 nearest-neighbor connectivity; edges are validated by collision sampling at resolution 0.010 of the side length, and worlds whose roadmap fails to connect start to goal are rejected and resampled.

Dynamic data collection. A candidate training world is *usable* iff (i) the world builds, (ii) its roadmap connects start to goal, and (iii) the space–time problem is solvable from the start at t=0 (finite backward space–time Dijkstra value). Training worlds are drawn only from the dynamic maze, rooms, and spiral families (dense maze, crossing, and large rooms are family-level held out from dynamic training). The heavy run draws seeds until 53 usable worlds are collected; each world contributes its reachable (v, t) states (192 nodes across $t_{\max}$+1 time steps, masked to reachable), averaging 21,312 states per world for 1,129,536 supervised scalar samples in total. All three training seeds use the same collection procedure with independent seeds.

B. Full Static and Dynamic Suite Tables

Tables 6 and 10 give the complete per-suite C7 static and C8 dynamic results summarized in the main paper; Tables 7–9 give the fixed-provider primary view and its replications.

Multi-seed replication detail (Table 9). Two full-pipeline retrains (collect + train, seeds 2001/2002) of the aware/blind field U-Net twins were run under a design frozen before training, then evaluated on the same frozen fresh cohort with the canonical binding budgets. Matched-ratio medians across seeds: maze 0.047–0.059, rooms 0.117–0.135, spiral 0.065–0.092 (seed-stable); crossing 0.273–0.434, dense maze 0.216–0.625, large rooms 0.250–0.548 (seed-variable). Preregistered readouts: R1 (success CI excludes zero in ≥5/6 suites) 5/6 and 6/6—pass; R2 (deltas within ± 0.15 of canonical in ≥5 suites) 5/6 and 6/6—pass; R3 (no suite significantly aware-over-blind) fails as stated for seed 2001 (large rooms $+0.080$ $[+0.020, +0.160]$)—the same suite that is significantly *blind*-better under seed 1234 (-0.200 $[-0.320, -0.100]$). Across the 18 seed–suite twin contrasts: 2 significantly aware, 2 significantly blind, 14 null, with the significant cells in different suites for different seeds; the per-suite future-window effects flip sign across seeds and are consistent with training-seed noise. No retraining, reselection, or recalibration occurred in response to any readout.

Success as a function of budget (Figure 3). A single evaluation pass at $4\times$ the binding budget on the same frozen cohort yields *exact* success-vs-budget curves for every lower budget: the space–time A* expansion order is budget-independent, and a map is solved at budget B iff its recorded solve expansions are $\leq B$, so the curves are derived by thresholding with no re-runs and no interpolation (design frozen before launch). As a determinism cross-check, thresholding at the binding budget reproduces every fresh-cohort success value exactly (all suites, all providers). The headline result is not an artifact of the calibrated operating point: Euclidean search needs 1.7–2.8× the binding budget to reach the fixed blind provider's binding-budget success (crossing 2.49×, maze 2.29×, dense maze 1.95×, rooms 1.73×, large rooms 1.94×, spiral 2.80×), and by $4\times$ the binding budget every provider converges to the same feasibility ceiling (0.94–1.00; on dense maze the 0.94 ceiling binds even the oracle, reflecting maps unsolvable within the time horizon).

C. Transfer and Composition Details

Grain note: the point values in this section are the original record-level curve summaries, retained for continuity with the run reports; the map-clustered reanalysis in Section J supplies the primary inferential values quoted in the main

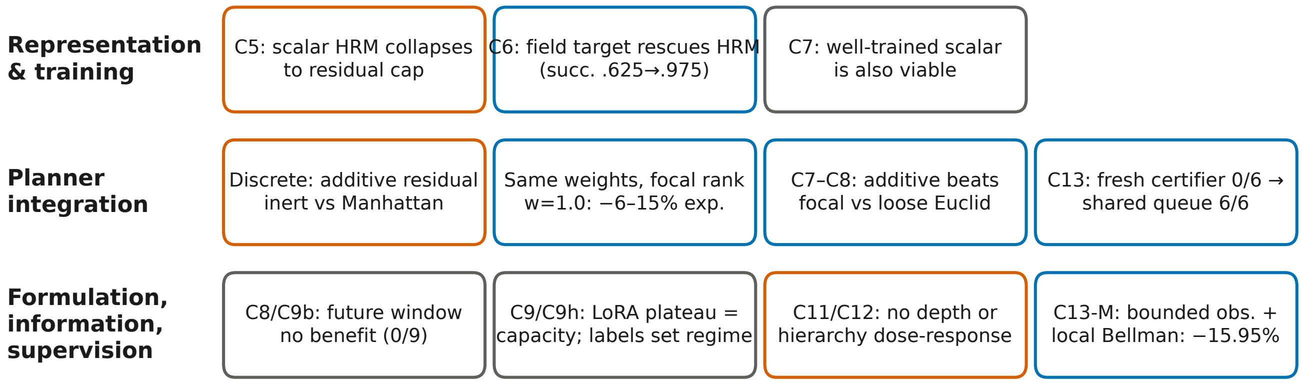

Figure 1: Program map: the three factors the study isolates—training formulation, planner integration, and information/supervision—with the experiments that intervened at each and their verdicts (orange: failure or negative; blue: positive under the corrected formulation or integration; gray: mechanism or null result).

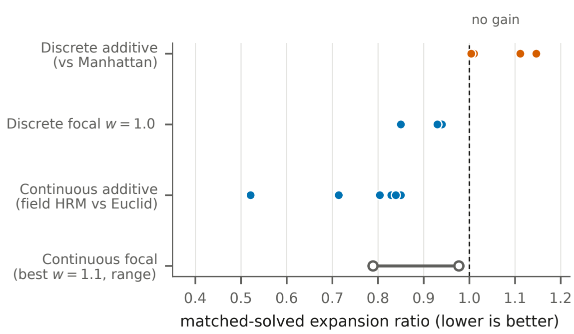

Figure 2: Integration summary across substrates (moved from the main paper). Discrete grids (Manhattan anchor): additive learned magnitude is at or below baseline while the same trained predictions reduce expansions by 6–15% as a within-band ranker at w=1.0. Continuous roadmaps (Euclidean anchor): additive guidance reduces matched expansions by 15–48%; the plotted focal w=1.1 range is an exploratory, post-hoc operating point and is not used for any main-paper claim.

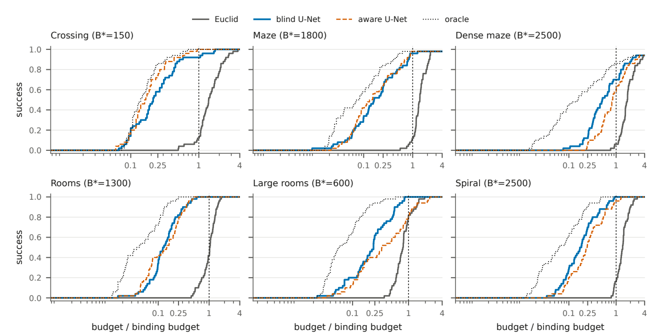

Figure 3: Success-vs-budget step curves per dynamic suite (fresh 50-map cohort), derived exactly from one 4×-binding evaluation pass. Budgets are normalized by the binding budget B^* (dashed vertical line). The learned providers reach high success at 0.05–0.3× B^*; Euclid requires 1.7–2.8× B^* to match the blind provider's binding-budget success.

paper, and where the two differ (e.g., C9h exact-zero counts, C9b's single positive cell) the map-level values govern.

Representative C9 HRM curves (ratio at success): maze-dense LoRA 0.650 (1.00) at K=1 to 0.730 (0.97) at K=16; maze-dense full FT 0.744 (0.76) to 0.571 (0.99) at K=16 and 0.552 at K=32; scratch 1.008 (0.37) to 0.808 (0.80); rooms-large full FT 1.128 (0.37) at K=1 to 0.500 (0.92) at K=8; bugtrap LoRA degrades from ≈0.696 (0.83) to 0.950 (0.48) at K=32—LoRA is usually but not universally base-preserving. C9H (matched compute): bounded-minus-unbounded LoRA median ratio delta 0.000 ± 0.008 across 27 cells (22/27 exactly zero; max $|\Delta|$ 0.028); rooms-large field U-Net full FT reaches 0.404 at 0.97 success from 0.982 zero-shot while conv-LoRA stays at 0.992–1.002. C9B (dynamics): maze-dense aware U-Net zero-shot 0.145 (0.60), LoRA K=16 0.059 (0.97), full FT K=1 0.112 (0.70); aware-minus-blind success deltas at full-FT K=16 are 0.00 or −0.05 in all nine cells. C10: RBF-merge minus zero-shot expansion-ratio deltas by target/backbone: −0.022, −0.012, −0.024, +0.082, +0.012, +0.116; weight-space vs. prediction-space deltas lie within $[-0.012, +0.007]$; own-axis RBF mass 0.986–0.998.

D. C11 Gates, Collapse Forensics, and Scale

Collapse forensics. Five of 33 recurrent/ACT arm-cells show constant-prediction or partial seed collapse (HRM-trace A8/C8, ON-LSTM B0, HRM-v2 B0, two ON-LSTM A8 seeds), with a reproducible loss signature at the normalization cap. The largest recurrent deficits (HRM-trace ratios 1.526 at A8, 1.600 at C8) are collapse measurements, not expressiveness estimates. Removing K=0 padding *worsens* recurrent predictions: models used pad steps as extra recurrent compute.

Scaled addendum. A 12-run `unet_film_big` vs.

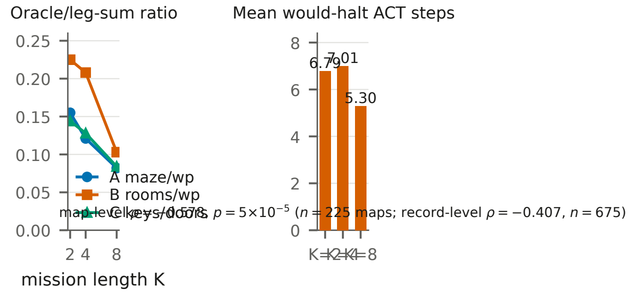

Figure 4: Compositional-mission hierarchy test (C11); K here is mission length. Left: the exact oracle's ratio over a leg-sum baseline falls as missions deepen (headroom exists). Right: learned HRM halting uses *fewer* steps at K=8; the annotation reports both analysis grains (map-level $\rho = -0.578$, $p = 5 \times 10^{-5}$, n=225 maps; record-level $\rho = -0.407$, n=675).

`hrm_trace_big` comparison (complete; 2,700 eval rows) preserves the ordering: completion 0.804 vs. 0.736 at K=2 and 0.413 vs. 0.307 at K=8; mean state MAE 0.395 vs. 0.447 (K=2) and 0.906 vs. 1.855 (K=8). At K=8 the scaled HRM exactly matches the leg-sum baseline's completion and mean expansions (0.307; 701.34)—persistent collapse, not recovered hierarchy.

E. C12 Detail

C12-A (memory headroom, then learned pilot): constructed decision-relevant aliasing at 71.3% of paired decisions; privileged mode/history diagnostic +0.467 completion and −65.1% collision-adjusted regret (map-bootstrap 95% CI 63.0–66.9%); oracle completion 0.975. The one-seed matched temporal-hierarchy pilot fails G1 (forecast), G2 (planning), and G3 (persistent carry); overall *strong negative*. C12-B (tied refiner): cycle-1→8 normalized burden improves monotonically in every cell (A/$K2$ 0.859→0.815; A/$K8$ 0.624→0.576; C/$K2$ 0.867→0.777; C/$K8$ 0.576→0.517; all cycle-1/8 map-bootstrap intervals separate), but the $K8-K2$ gain is $+0.0038$ (CI $[-0.0199, 0.0281]$) on A and -0.0305 ($[-0.0726, 0.0096]$) on C—no dose–response. Tied beats shallow and untied on C/$K8$ (deltas 0.0177/0.0076, BH q=6.7×10^{-5}) but loses to untied on A/$K8$ (q=0.0168). Bellman residual *worsens* over cycles in every cell while burden improves; solved cycle-8 paths average 1.4–2.2% above oracle cost. K=16 was dropped before training by a construction gate (only 2/300 and 1/300 valid missions).

F. C13: Mechanism Ladder and Frozen Confirmation

Claim supported. A provider restricted to bounded local observations (radius-0.20 subgraph on a known roadmap; behavior-derived labels; no shortest-path supervision) can, with one inference-time local Bellman backup, beat every fixed complete-map C7 provider in pooled expansions at matched empirical path quality on a frozen, disjoint cohort.

Ladder. Each rung changes one factor; deltas are paired expansions versus the matched comparator (rungs C–G versus matched focal search at w=1.10, including oracle providers on six development maps; rungs I–M versus the complete-map field HRM, live on all six suites).

Frozen confirmation, per suite. 144 maps (24 per family; seeds disjoint from every prior cohort; feature caches hash-verified against live regeneration; model, radius 0.20, α=1.50, integration, and analysis frozen before generation).

Scale calibration. On the disjoint 48-map calibration block, $\alpha \in \{1.00, 1.25, 1.50, 2.00\}$ gives deltas -3.79 $[-9.40, +1.73]$, -9.77 $[-15.38, -4.42]$, -13.04 $[-18.67, -7.65]$, and -16.94 $[-22.56, -11.54]$, with 4/4/6/6 negative suite means and mean/max cost ratios rising from 1.022/1.241 to 1.052/1.366; α=1.50 was frozen as the largest scale whose suite breadth was complete before the absolute-ceiling verdict, and every arm formally fails the preregistered absolute 1.10 maximum-cost ceiling. *Chronology, stated explicitly:* the absolute ceiling was part of the original calibration design; its failure was observed on the calibration block, during development and before any confirmation data existed; the comparator-relative criterion (mean path-cost ratio within +0.005 and maximum within +0.02 of the complete-map field HRM) and the α=1.50 selection were then written into the frozen confirmation design *before* the 144-map cohort was generated, and no criterion changed afterward. Under that gate the confirmation passes with margin (1.0235/1.1624 vs. 1.0311/1.3346); had the original absolute ceiling been retained, the direct arm would fail it on maximum cost, which we report rather than suppress.

Operating points and timing. The direct α=1.50 arm is the matched-quality expansion result and carries no formal bound. The separate certified operating point runs the learned rank inside a w=1.10 focal band with a shared-queue consistent-bound incumbent proof and passes all 144 bound and certificate checks. Per-map wall time decomposes as feature construction 5.138 s (prototype Python builder), model inference 0.0008 s, local backup 0.0097 s, and search 0.0002 s, versus 0.371 s complete-map field-HRM inference: the result is search effort and path quality, not latency.

Architecture substitutions under the identical method (C13-N/O/P) are summarized in the main paper; their raw artifacts (development grids, verdicts, integrity manifests) ship in the artifact package.

G. Discrete Program Detail

The discrete program is substantially larger than the canonical subset reported here: fifteen experiment families across eleven surveyed cloud volumes (13,671 parsed result files) and roughly thirty experiment scripts, including a ten-run early scaling lineage of obstacle-trajectory forecasters, diffusion path-generator baselines, a superseded harder-calibration predecessor of the structured protocol below, partial precursors of the controlled transfer run, an empty-model baseline-rerun control (static A* re-verified through the learned-model code path), and a budget-invariance analysis establishing that raising search budgets adds work without rescuing success on the hard families. The paragraphs below report the completed canonical results; the full inventory ships with the released artifacts.

Forecasting. Matched 100-episode protocol: HRM-3M/10M 71/100 vs. LSTM 67–69/100 (no uncertainty reported; two-episode matched gap). Structured Preset M+ (four 100-episode suites): ON-LSTM-10M 0.3875 mean success vs. HRM-10M 0.2650 and HRM-3M 0.200; ON-LSTM-3M also has the lowest rollout MSE (0.1257/1.1269 one-/five-step vs. 0.3973/2.6438 for HRM-3M).

Additive transfer (clean v3, 22 suites $\times$ 3 budgets). Manhattan 0.590 success / 126,707 mean expansions; HRM full-FT 0.585 / 128,013; ON-LSTM full-FT 0.564 / 127,228; HRM LoRA 0.554 / 140,852; ON-LSTM LoRA 0.514 / 145,282. Every learned arm $\leq$ matched baseline.

Pooled multitask. HRM pooled base 0.612 success (+2.11pp) / 122,184 expansions vs. Manhattan 0.591 / 126,630—the strongest completed discrete additive positive; only 1 of several specialists beats its matched four-suite baseline (+2.0pp; others -0.42 to -20.25pp).

Focal ranking (local pilot, 3–8 seeds). Magnitude is scale-flat despite near-perfect ordering (correlation 0.987–0.994; predicted/true mean 0.73/0.42/0.27/0.19 at 64/128/192/256). At w=1.0: HRM A128-static ratio 0.85 with success 0.62→0.75; A192 0.94 (0.75→0.75); A128-dynamic 0.93 (0.62→0.62); ON-LSTM A128/A192 0.85 (0.75→0.75). At w=1.05, A128-dynamic improves further but A192 success regresses 0.75→0.62; w=1.0 is the evidence-safe operating point. Experts exactly match pooled bases in tested cells.

H. Validity Forensics

The frozen-halt artifact. A 27.27M-parameter direct maze solver reported 96.64% token accuracy, 25.40% exact accuracy, and "74.60% halt accuracy" with 16 average adaptive-computation steps. Audit: $74.60 = 100 - 25.40$ exactly—the halt logit was frozen negative (a constant-false classifier) because the training loop omitted the halt loss and back-propagated only the final segment. After repairing the loss path (verified live: halt loss 0.14→0.06, token accuracy 90–95%), exact accuracy remained 0.000 at the chosen stop point; the repaired run demonstrates mechanism liveness, not task convergence. Bit-exact forward parity with the reference implementation was separately established. We recommend the $100-x$ pattern as a standard red flag for adaptive-computation metrics.

The cap-collapse signature. C5's scalar HRM collapse sits at a loss value pinned by the residual cap (2.612 in that setup, with constant predictions at the cap). The same class of signature reappears in C11: collapsed cells share an *identical* final loss (1.1397, exactly the constant-cap loss under that normalization), enabling automatic collapse detection instead of misreading collapsed cells as capacity evidence.

Pad steps as compute. In C11, removing K=0 padding tokens *degraded* recurrent predictions—models had co-opted padding as additional recurrent iterations. Sequence-model comparisons that alter padding alter compute.

I. Statistics

Success is compared with paired McNemar tests, BH-corrected within stage families. Expansion effort is compared only on matched solved maps via median ratios with bootstrap percentile CIs (10k–20k resamples); from C12 onward, bootstraps are map-clustered by design. C9–C11 originally pooled repeated evaluations of fixed test maps across adaptation/model seeds; Section J reports the map-clustered reanalysis of every quantity the main paper quotes from those stages; the fixed-provider dynamic reanalysis of Table 7 uses the same map-level grain. Matched-solved ratios condition on joint success and can favor arms that solve only easier maps; we therefore always report success and ratios separately and never combine them into a single score. Negative results are failures to observe an effect under the tested protocol, not equivalence claims; no equivalence tests were run.

J. Map-Clustered Reanalysis

Units are test maps; adaptation/model seeds are averaged within maps before inference; success deltas use 10,000-resample map bootstraps; expansion ratios are medians of per-map ratios over matched maps at the binding budget in additive mode; the C11 halting test aggregates model seeds within maps and permutes mission length within config (maps are independent across K cells; 20,000 stratified permutations). Estimator conventions differ slightly from the published record-level curves, so point values differ modestly; the question is verdict-level agreement. A methodological caution surfaced during this work: a first pass that accidentally pooled additive and focal evaluation modes attenuated the apparent C9 transfer effects by roughly half—mode/aggregation mixing alone can manufacture or destroy an effect of the size under study.

C11 halting. Record level reproduces the published value ($\rho = -0.4066$, n=675); map level strengthens it ($\rho = -0.5778$, n=225, stratified permutation p=5×10^{-5}). Map-level means 6.78/7.01/5.30 at K=2/4/8.

C11 vs-MLP ratios. Map-clustered ratio-of-means CIs exclude parity at $K \in \{0, 2\}$ in config A for both U-Net (0.734/0.846) and GNN (0.822/0.923), and in config B at K=0 (0.904/0.818) plus B/K=2 for the GNN (0.805); config C never separates; every $K \in \{4, 8\}$ cell includes parity.

C9H bound-vs-capacity. 27 cells: median bounded-minus-unbounded map-level ratio delta +0.0000; 15/27 exactly zero; sd 0.0077; max $|\Delta|$ 0.0370—the capacity attribution stands.

C9B aware−blind (full FT, K=16). Eight of nine point deltas are zero or negative; the single positive (+0.017 $[0.000, +0.050]$) does not exclude zero; no cell shows a significant aware advantage.

C10 composition. Map-paired deltas vs. zero-shot: four cells ≈0 with CIs crossing zero; the two ON-LSTM rooms cells are significantly *worse* ($+0.089$ $[+0.061, +0.121]$ and $+0.098$ $[+0.076, +0.123]$), matching the published point deltas and upgrading the null to "never better, sometimes significantly worse."

No preregistered verdict flips under any of these recomputations.

J1. Tuned Weighted-A* Control

J2. Fixed-Model Selection Audit Trail

The primary dynamic claim rests on one fixed learned heuristic; its provenance is as follows. (1) *Candidates:* the original dynamic run trained aware and blind twins of scalar HRM, scalar ON-LSTM, and field U-Net models. (2) *Selection:* the field U-Net blind twin was designated the single primary model for every suite during the fixed-model reanalysis design; the original 20-map evaluation results were known at that time, so the accurate statement is that the model, integration mode (additive), and per-suite binding budgets were **fixed before the 50-map confirmation cohort was generated**, not that selection predates all target outcomes. (3) *Freeze:* the confirmation cohort's generation seed (999,999) was chosen in advance; the evaluation-seed formula's within-suite offset bound guarantees map-disjointness from every earlier cohort, and world fingerprints (start, goal, obstacle coordinates) were checked. (4) *Integrity:* the fixed checkpoint's SHA-256 is `b8378950545f8abdbf06d59568b5d8ab6069884c1c038d23a41a14b6dc17fb6f`; the binding-budget calibration file is copied verbatim into every evaluation directory. (5) *Seeds:* the two additional training pipelines change only the training seed (2001, 2002); collection, architecture, integration, budgets, and the evaluation cohort are identical.

K. Reproducibility and Artifact Inventory

From C7 onward every stage freezes a written design (gates, budgets, analysis grain) before execution; artifacts (checkpoints, feature caches, cohort manifests, raw rows, verdicts, reports) are hash-bound in integrity manifests; confirmation cohorts are generated only after freezing model, integration, and evaluation code; failed development gates block confirmation rather than triggering retuning. Duplicate-evaluation determinism is enforced where preregistered (byte-identical traces; identical decision rows). The final-stage pilot on persistent planning-time state ran under a fully test-verified harness (158 focused-plus-neighboring tests; hardware preflight). Its first staged execution ended at a preregistered execution-integrity hard stop—the harness's independent reanalysis disagreed with the primary comparison computation, voiding all mechanism fields. After repair of the named defect only, a fresh-fingerprint rerun passed every integrity primitive, including independent reconstruction of the comparison evidence from promoted raw artifacts, and returned the valid negative reported in the main paper: persistent carry loses to the same-checkpoint reset twin on map-macro frontier ranking (MRR delta -0.029, 95% CI $[-0.060, -0.003]$), and no self-bootstrap or confirmation was run. We note the initial stop as the integrity machinery functioning as designed.

The anonymized artifact archive submitted with this paper contains: (i) the analysis scripts that regenerate every main-paper figure and quoted map-level statistic from raw rows (the fixed-provider dynamic reanalysis, the map-clustered reanalysis, and all figure generators); (ii) the raw paired evaluation rows behind every quoted quantity (C7/C8 evaluation tables, including the fresh-cohort and three-seed replication rows, C9/C9H/C9B/C10 evaluation tables, C11 evaluation and halting tables, and C13 development and confirmation rows); (iii) map/cohort manifests, seed lists, frozen configurations, and the integrity manifests for the frozen confirmations; (iv) the preregistration design documents from C7 onward; and (v) a top-level README with reproduction commands and expected outputs. Checkpoints are included where size permits; otherwise exact training commands and configurations are provided.

L. C14: Label-Count × World-Diversity Factorial

A preregistered factorial (design and pre-execution amendment in the artifact package) tests whether the LoRA/full-fine-tuning contrast is governed by the number of supervised search states N, independent of domain and map count. Factors: $N \in \{256, 1024, 4096, 16384, 65536\}$ exact-count state samples × two diversity levels at fixed N (concentrated: the minimal world count $w_{\min}(N)$; distributed: $8 \times w_{\min}(N)$ worlds) × two domains (static C9 substrate and dynamic C9B substrate, maze-dense target, frozen sources) × {unbounded rank-8 LoRA, full fine-tuning, scratch} × 3 seeds — 180 adaptations, every cell at an identical 2,560 optimizer steps, evaluated on the frozen C9/C9B test cohorts at binding budgets. The measured $w_{\min}$ spans quantify the domains' supervision densities: reaching N=65,536 takes 356 static worlds but 4 dynamic worlds (∼185 vs. ∼18,900 usable states per world).

Preregistered verdict: H-C14 rejected as stated. On matched-solved expansion ratios the hypothesized crossover point is degenerate: full fine-tuning is at or below LoRA at *every* tested N in all 12 (domain × diversity × seed) cells, and the preregistered regression's full-FT × $\log N$ interaction is null (-0.0025 $[-0.0087, +0.0028]$). State count improves ratios ($\log_2 N$ coefficient -0.0140 $[-0.0190, -0.0081]$) but identically for both methods.

The preregistered diversity readout localizes the real effect. The low-supervision protection LoRA provides appears in *success*, and it is governed by world diversity at fixed N, not by state count. Concentrated low-world cells collapse full fine-tuning and scratch in both domains (success vs. the frozen zero-shot source: -0.300 $[-0.550, -0.050]$ in all three seeds at dynamic concentrated N=256 and N=1,024; -0.167 to -0.300 at static concentrated N=256), while the distributed cells at the same N rescue them (dynamic $+0.150$ to $+0.200$; static ≈ 0.000; paired dist−conc full-FT deltas $+0.30$ to $+0.48$ dynamic, $+0.233$ static at N=256). LoRA never collapses in any of the 60 cells. The dynamic concentrated arm recovers exactly where $w_{\min}$ forces multiple worlds (N=65,536, 4 worlds: $+0.100$ in all seeds), tying the recovery to world count at fixed steps. This refines, without contradicting, the main text's supervision-density account: in the K-indexed adaptation protocols map count and state count move together, and C14 separates them — the protective variable is diversity (or adapter-restricted capacity), not raw state count.

Protocol detail. Measured minimal world counts

$w_{\min}(N)$ for N=256/1,024/4,096/16,384/65,536: static concentrated 2/6/22/89/356 worlds (distributed 8×: 16/48/176/712/2,848); dynamic concentrated 1/1/1/1/4 (distributed 8/8/8/8/32). Every adaptation runs exactly 2,560 optimizer steps—the epoch-equivalent of the largest N under the canonical recipe—so compute cannot proxy for labels; smaller-N cells cycle their fixed sample. The three adaptation seeds vary initialization and batch order only; the sampled state indices are shared across methods and seeds within a cell. *Collapse* denotes a cell whose adapted success falls significantly below the trained source model's zero-shot success (paired map CI excluding zero in the harmful direction). Inference unit: evaluation maps, seeds averaged within maps.

Limitations. Euclid solves 1/20 dense dynamic test maps at the binding budget, so dynamic matched-ratio cells have n=1 and carry no map-level uncertainty; dynamic inference rests on the 20-map paired success deltas. One target family per domain; three adaptation seeds; adaptations ran on cloud L4 hardware from locally collected, hash-recorded datasets.

Stage	Protocol essentials
C1–C4	9 easy suites, n=80/40 maps, budgets 100–200; pooled bases, TaskLoRA experts, RBF mixtures.
C5	calibrated hard maze/dense/rooms; 160 train / ∼80 eval maps; budgets 128–168; scalar residual target with cap.
C6	goal-conditioned raster residual field (8 input channels), sampled at PRM nodes; 96/40 maps; budgets 128–168.
C7	6 suites (3 held-out families); 96/24 maps per suite; binding budgets 140/24/56; scalar+field × additive+focal matrix; exact-Dijkstra oracle.
C8	deterministic moving circular obstacles; space–time PRM; aware (W=8 future window) vs. blind (W=0) twins; 53 training worlds (maze/rooms/spiral families); 1,129,536 scalar samples; 20-map/suite *development* cohort, then a 50-map/suite *confirmation* cohort evaluated after freezing, plus two independent retrained seeds on the same confirmation cohort.
C9/C9H	adapt C7 pooled bases to 3 held-out families; $K \in \{0..32\}$ labeled maps; LoRA(r8, bounded/unbounded), full FT, scratch; C9H matches compute (10 epochs, LR 2×10^{-4}); 30 fixed test maps/target.
C9B	same protocol on C8 dynamic sources (aware+blind), 486 adapters, 3 targets, $K \in \{1, 4, 16\}$.
C10	16 rank-8 experts on 2 family axes; zero-label transfer to 3 interior targets; RBF/nearest/uniform weight-merge and prediction-mix composition.
C11	compositional missions (waypoint sequences; key/door), configs A/B/C, mission length $K \in \{0, 2, 4, 8\}$; 40 train / 25 test maps per cell; 6 learned arms × 3 seeds (198 checkpoints; 54,450 eval rows); exact mission oracle and leg-sum baseline; forced-compute and would-halt probes.
C12	(A) hidden slow/fast regime dynamics, memory-headroom gate, matched temporal-hierarchy pilot; (B) tied vs. untied product-graph refiner, cycles 1–8, configs A/C × $K \in \{2, 8\}$, 3 seeds, 48 checkpoints, two 3,200-row test tables.
C13	current-state heuristic under bounded observation (radius 0.20 local subgraph); behavior-derived fresh-start rollout labels (median of 3); staged falsification ladder (A–L) then frozen 144-map confirmation (M); architecture substitutions (N/O); persistent-state pilot (P).
Discrete	20×20 to 256×256 grids; Manhattan-baseline A*; forecasting (dynamic obstacles), additive residual transfer (22 suites × 3 budgets × 100 episodes), pooled multi-task bases, learned focal ranking at $w \in \{1.0, 1.05, 1.1\}$.

Table 1: Protocol essentials by stage.

Family	Generator rule (lengths as fractions of side length)
Maze / dense maze	Corridor maze of axis-aligned walls; the dense variant narrows gap width to 0.115. Held out: dense.
Rooms / large rooms	2×2 rooms from one vertical + one horizontal partition wall, one doorway per wall, doorways biased toward opposite edges; large variant doubles side length. Held out: large.
Spiral	Four serpentine vertical walls spanning $x \in [0.22, 0.78]$, gap side alternating per wall. Trained.
Bugtrap	Solid vertical back wall, two horizontal arms, and a mouth lip closing the center of the opening (U-pocket). Held out.
Crossing (dyn.)	Open arena; dynamic timing control.

Table 2: Static geometry per family. Base-anchor families (open/clutter/narrow variants) used by the early stages additionally place 3–68 circular obstacles with radii 0.022–0.115; full ranges ship with the generator in the artifact package.

Suite	Patr.	Radius	Span	Period	v_{ag}	$t_{\max}$
Maze	3	0.075	0.38	0.14	0.060	110
Rooms	3	0.075	0.40	0.14	0.060	110
Spiral	3	0.075	0.38	0.14	0.060	110
Dense maze	4	0.062	0.42	0.17	0.060	140
Crossing	4	0.055	0.50	0.30	0.060	110
Large rooms	5	0.075	0.36	0.14	0.120	130

Table 3: Dynamic-suite parameters. Each patroller is a circle sweeping a line segment with triangle-wave motion; radius and span are fractions of the side length, period is a fraction of the horizon $t_{\max}\,dt$ (dt=1 s), and each patroller crosses its span twice per period. Sweep centers carry lateral jitter ≤ 0.02 (0.10 on crossing). The planner and collision checker always use these exact trajectories; only the *aware* heuristic input sees the W=8 future-occupancy channels.

Model	Configuration	Input	Parameters
Scalar HRM	hid. 192, 2 layers, 4 heads, k=2	62 tok. × 16	2,158,529
Scalar ON-LSTM	hid. 256, 2 layers, chunk 8	62 tok. × 16	1,252,481
Field U-Net	FiLM-conditioned, 8 input channels	64×64 raster	3,702,370
Field HRM	8 input channels	64×64 raster	4,016,642
Field ON-LSTM	8 input channels	64×64 raster	1,448,578

Table 4: Instantiated architectures with parameter counts computed from the released constructors. Scalar token sequences: one summary token, 12 nearest-obstacle tokens, 48 ray tokens, one goal token. Dynamic variants append W future-occupancy channels (W=8 aware; W=0 blind). LoRA inserts rank-8 adapters on the HRM (rank-24 on the ON-LSTM, matching its wider hidden layer); the quoted crossover cells are HRM (rank 8). Optimizer for all models: AdamW, learning rate 2×10^{-4} (1.5×10^{-4} for adapters), weight decay 10^{-4}, batch 256, gradient clip 1.0, smooth-L1 loss on the capped residual (cap 4.0), 10 base epochs (8 adapter epochs) unless a stage states otherwise.

Cell	K	Δsucc. [CI]	Δratio [CI]	n
dense/HRM	1	$-.240$ [$-.360, -.133$]	$+.112$ [$+.052, +.171$]	10
dense/HRM	16	$+.027$ [$-.020, +.100$]	$-.149$ [$-.179, -.117$]	10
dense/ON-L.	1	$-.047$ [$-.147, +.073$]	$-.162$ [$-.223, -.089$]	10
dense/ON-L.	16	$+.000$ [$-.027, +.033$]	$-.126$ [$-.171, -.083$]	10
bug./HRM	1	$-.467$ [$-.587, -.340$]	$+.225$ [$+.089, +.352$]	13
bug./HRM	16	$-.020$ [$-.120, +.080$]	$-.145$ [$-.239, -.043$]	16
bug./ON-L.	1	$-.093$ [$-.207, +.007$]	$+.229$ [$+.117, +.348$]	15
bug./ON-L.	16	$+.100$ [$+.013, +.207$]	$-.052$ [$-.154, +.033$]	15
rl./HRM	1	$-.600$ [$-.773, -.407$]	$+.290$ [$+.207, +.360$]	11
rl./HRM	16	$+.073$ [$-.027, +.187$]	$-.366$ [$-.489, -.235$]	11
rl./ON-L.	1	$-.293$ [$-.433, -.160$]	$+.013$ [$-.156, +.167$]	11
rl./ON-L.	16	$-.053$ [$-.167, +.053$]	$-.170$ [$-.326, -.010$]	11

Table 5: Direct map-paired full-FT − LoRA contrasts (dense = dense maze, bug. = bugtrap, rl. = large rooms; ON-L. = ON-LSTM). Adaptation seeds are averaged within maps; intervals are 10k map-clustered bootstraps over the 30 test maps per target; Δratio is computed on the n triple-matched maps solved by both methods and the anchor, as a difference of expansion ratios relative to the anchor (negative favors full fine-tuning). At K=1 the success contrast favors LoRA in all six cells (four significantly); at K=16 the effort contrast favors full fine-tuning in all six cells (five significantly) at near-parity success.

Suite	Euclid succ.	Field-HRM succ.	Ratio [95% CI]
Maze	0.583	1.000	0.521 [0.450, 0.627]
Maze-dense	0.250	0.958	0.804 [0.707, 0.829]
Rooms	0.375	0.958	0.829 [0.775, 0.885]
Spiral	0.250	0.917	0.850 [0.742, 0.919]
Bugtrap	0.458	0.750	0.714 [0.533, 0.800]
Rooms-large	0.417	0.750	0.839 [0.646, 1.222]

Table 6: C7 static suites at binding budgets: success and matched-solved median expansion ratio (field HRM vs. Euclid). Success gains are BH-significant in 4/6 suites (maze, dense, rooms, spiral); bugtrap and rooms-large reach corrected significance only for other learned arms. Scalar-HRM ratios span 0.427–0.822. Matched n ranges 6–14.

Suite	Euclid succ.	Blind U-Net succ.	Ratio [95% CI]	n
Crossing	0.30	0.95	0.191 [0.166, 0.878]	6
Maze	0.30	1.00	0.093 [0.057, 0.137]	6
Dense maze	0.05	0.75	0.246 (descriptive)	1
Rooms	0.30	1.00	0.099 [0.050, 0.214]	6
Large rooms	0.75	0.95	0.228 [0.185, 0.418]	15
Spiral	0.20	0.95	0.087 [0.061, 0.287]	4

Table 7: C8 fixed-provider result on the *original* 20-map cohort (field U-Net blind; no future-motion input; additive mode; binding budgets; map-level bootstraps). Success deltas are $+0.20$ to $+0.75$ with all six paired CIs excluding zero.

Suite	Euclid succ.	Blind U-Net succ.	Ratio [95% CI]	n
Crossing	0.12	0.92	0.273 [0.109, 0.516]	6
Maze	0.12	0.96	0.047 [0.021, 0.153]	6
Dense maze	0.06	0.70	0.216 [0.100, 0.281]	3
Rooms	0.42	1.00	0.121 [0.106, 0.172]	21
Large rooms	0.82	1.00	0.250 [0.188, 0.306]	41
Spiral	0.16	1.00	0.092 [0.065, 0.146]	8

Table 8: C8 fixed-provider *replication* on the fresh 50-map-per-suite cohort (generation seed 999999, frozen before any result was observed; map-disjoint from the original cohort by the evaluation seed formula's within-suite offset bound). Success deltas are $+0.18$ to $+0.84$ with all six paired CIs excluding zero. Map-paired aware$-$blind success deltas on this cohort: crossing $+0.080$ $[+0.020, +0.160]$, maze $+0.020$ $[0.000, +0.060]$, dense maze -0.120 $[-0.220, -0.040]$, rooms 0.000, large rooms -0.200 $[-0.320, -0.100]$, spiral -0.040 $[-0.100, 0.000]$: one suite significantly favors the aware twin, two the blind twin, three neither.

Suite	Seed 1234	Seed 2001	Seed 2002
Crossing	$+0.80$ [0.68, 0.90]	$+0.86$ [0.74, 0.96]	$+0.88$ [0.78, 0.96]
Maze	$+0.84$ [0.74, 0.94]	$+0.86$ [0.76, 0.96]	$+0.84$ [0.74, 0.94]
Dense maze	$+0.64$ [0.50, 0.78]	$+0.46$ [0.32, 0.60]	$+0.56$ [0.42, 0.70]
Rooms	$+0.58$ [0.44, 0.72]	$+0.58$ [0.44, 0.72]	$+0.58$ [0.44, 0.72]
Large rooms	$+0.18$ [0.08, 0.30]	$+0.10$ [-0.02, 0.22]	$+0.16$ [0.06, 0.26]
Spiral	$+0.84$ [0.74, 0.94]	$+0.84$ [0.74, 0.94]	$+0.84$ [0.74, 0.94]

Table 9: C8 preregistered multi-seed replication: paired success deltas (blind U-Net $-$ Euclid, map-level 95% CIs) for three independent training seeds on the same frozen fresh cohort. 17 of 18 CIs exclude zero; the single crossing interval is seed 2001's near-ceiling large-rooms cell (Euclid 0.82).

Suite (arm)	Succ. E$\to$L	Ratio [95% CI]	Matched n
Crossing (U-Net)	0.30$\to$1.00	0.258 [0.132, 0.769]	6
Maze (U-Net)	0.30$\to$1.00	0.064 [0.046, 0.088]	6
Maze-dense (U-Net)	0.05$\to$0.75	0.278 (no CI)	1
Rooms (U-Net)	0.30$\to$1.00	0.096 [0.038, 0.191]	6
Rooms-large (U-Net)	0.75$\to$0.90	0.380 [0.132, 0.591]	14
Spiral (field HRM)	0.20$\to$0.90	0.046 [0.038, 0.073]	4

Table 10: C8 *secondary* view: per-suite strongest arms (explicitly post-hoc selection). Corrected success evidence holds on five suites; rooms-large is supported mainly by expansion evidence. Dense-maze has matched n=1 and supports no expansion inference. Blind arms are the per-suite best in five of six suites.

Gate	Verdict
K=0 continuity	FAIL, diagnosed as false premise: C7 had no MLP arm and did not give every architecture the C11 input; real shallow separation plus collapsed recurrent cells violate the all-tie assumption. Measurement layer verified.
G1 dose–response	Negative: no structured arm beats MLP at $\geq$2 mission depths with non-decreasing gap.
G2a forced compute	Negative: HRM-v2 k=$1/2/4/8$ expansion and MAE curves flat in every K=8 config.
G2b learned halting	Negative, inverted: map-level Spearman $\rho = -0.578$, stratified permutation p=5×10^{-5}, n=225 maps (record-level $\rho = -0.407$, n=675, the original gate value); mean would-halt steps 6.79/7.01/5.30 at K=$2/4/8$.
G3 closure	Neither preregistered branch: global-input arms separate at shallow K (U-Net/GNN vs. MLP ratios ≈0.67–0.87; state MAE 0.151/0.171 vs. 0.251 at A/K=0), but no arm converts depth into growing advantage.

Table 11: C11 preregistered gates.

Rung	Factor changed	Δ exp. [95% CI]
C	fresh certifier (oracle rank)	$+11.50$ (0/6 wins)
D	shared-queue certifier (oracle rank)	-6.83 (6/6 wins)
E	exact rollout rank	$+1.33$ (2/6)
F	monotone recalibration	$+1.33$
G	analytic local escape	$+6.83$ (0/6)
I	one-family model, live six suites	$+15.36$ $[+12.02, +18.64]$
J	suite-balanced training, static insert	$+16.17$ $[+9.67, +22.67]$
K	+ one local Bellman backup	-1.21 $[-8.04, +5.46]$
L	+ scale α=1.50 (48 fresh maps)	-13.04 $[-18.67, -7.65]$
M	frozen 144-map confirmation	-12.96 $[-16.30, -9.74]$

Table 12: The C13 falsification ladder. Target repairs (E, F), analytic constructions (G), and distribution repairs (I, J) fail; the integration repair (K) plus scale calibration (L) flip the sign, and the frozen confirmation (M) reproduces it.

Suite	Local	Field HRM	Δ	95% CI	W/T/L
Maze	47.63	84.38	-36.75	$[-45.58, -27.83]$	24/0/0
Dense maze	95.38	104.50	-9.13	$[-15.21, -3.13]$	18/1/5
Rooms	98.04	106.17	-8.13	$[-14.33, -1.71]$	16/0/8
Spiral	120.25	121.33	-1.08	$[-5.88, +4.21]$	14/1/9
Bugtrap	14.96	24.29	-9.33	$[-17.79, -2.92]$	18/1/5
Large rooms	33.58	46.92	-13.33	$[-18.92, -7.50]$	19/0/5
Pooled	68.31	81.26	-12.96	$[-16.30, -9.74]$	109/3/32

Table 13: C13-M per-suite paired expansion deltas versus the complete-map field-HRM comparator. Every suite mean is negative; the spiral interval crosses zero, and the preregistered gate is pooled with a suite-breadth condition (at least four negative suite means), which passes. The local provider also beats every other fixed C7 learned provider in pooled expansions on this cohort.

Cell (HRM)	Δsuccess [CI]	map ratio [CI]
dense LoRA $K{=}1$	$+0.467$ $[+0.300, +0.633]$	0.686 $[0.590, 0.774]$
dense LoRA $K{=}16$	$+0.433$ $[+0.267, +0.600]$	0.786 $[0.694, 0.891]$
dense full $K{=}1$	$+0.227$ $[+0.080, +0.380]$	0.789 $[0.751, 0.836]$
dense full $K{=}16$	$+0.460$ $[+0.287, +0.633]$	**0.610** $[0.549, 0.670]$
dense scratch $K{=}1$	-0.167 $[-0.300, -0.047]$	1.035 $[1.029, 1.039]$
rl. full $K{=}1$	-0.167 $[-0.340, +0.013]$	1.293 $[1.170, 1.503]$
rl. full $K{=}8$	$+0.387$ $[+0.227, +0.547]$	0.590 $[0.498, 0.712]$
bug. LoRA $K{=}32$	-0.320 $[-0.507, -0.133]$	1.153 $[1.012, 1.523]$

Table 14: Map-clustered C9 key cells (binding budgets, additive mode; dense = maze-dense, rl. = rooms-large, bug. = bugtrap). The $K{=}16$ full-FT/LoRA CIs are disjoint (the crossover), the rooms-large $K{=}1$ catastrophe excludes parity, and the bugtrap LoRA degradation is significant in both metrics.

Suite	w_h	WA*	Blind	Δsucc. [CI]	Bl./WA* (n)	Sub. W/B
Crossing	1.5	1.00	0.92	$-.08$ $[-.16, -.02]$	0.825 (46)	1.008/1.164
Maze	5	0.96	0.96	$+.00$ $[-.06, +.06]$	0.601 (47)	1.017/1.012
Dense maze	5	0.70	0.70	$+.00$ $[-.06, +.06]$	0.982 (34)	1.006/1.008
Rooms	3	0.98	1.00	$+.02$ $[+.00, +.06]$	0.554 (49)	1.015/1.019
Large rooms	1.5	0.98	1.00	$+.02$ $[+.00, +.06]$	0.728 (49)	1.003/1.082
Spiral	5	0.70	1.00	$+.30$ $[+.18, +.42]$	0.217 (35)	1.013/1.011

Table 15: Tuned weighted A* versus the fixed blind U-Net on the confirmation cohort at binding budgets. w_h is the per-suite inflation selected on the development cohort under the pre-registered rule (highest success, ties toward smaller w_h). Δsucc. is the paired blind−weighted success difference (10k map bootstraps); Blind/WA* is the median expansion ratio on the n jointly solved maps (below 1 = learned expands fewer nodes); sub. is mean empirical suboptimality (arrival over optimal arrival) on solved maps. The learned heuristic expands fewer nodes in every suite, wins spiral success outright, and loses crossing success to inflation; results reported as-is per the frozen design.